

Name: J. Renuka Devi

Reg.no: 192573007

Course code: DSA0602

Course name: Data Handling and Visualization

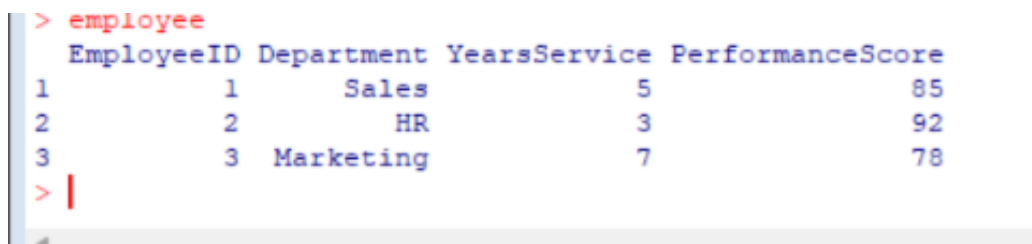
LAB PROGRAMS (8-12)

SET 8 EMPLOYEE PERFORMANCE ANALYSIS

Dataset: Employee Performance

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)
```

employee



```
> employee  
  EmployeeID Department YearsService PerformanceScore  
1           1     Sales             5              85  
2           2        HR             3              92  
3           3  Marketing             7              78  
> |
```

1. Create a line chart to visualize the performance trend of employees over time.

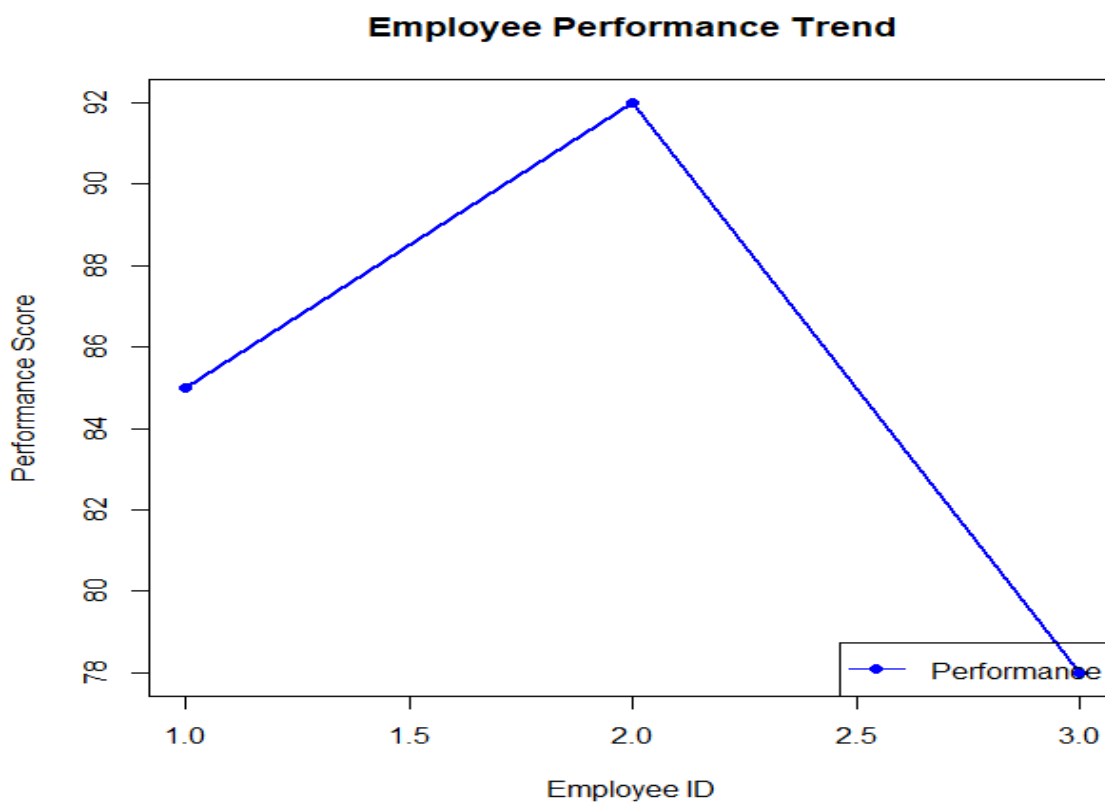
Label the axes and title the chart.

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)
```

```
plot(employee$EmployeeID,  
  employee$PerformanceScore,  
  type="o",  
  pch=16,  
  col="blue",  
  lwd=2,
```

```
xlab="Employee ID",  
ylab="Performance Score",  
main="Employee Performance Trend")
```

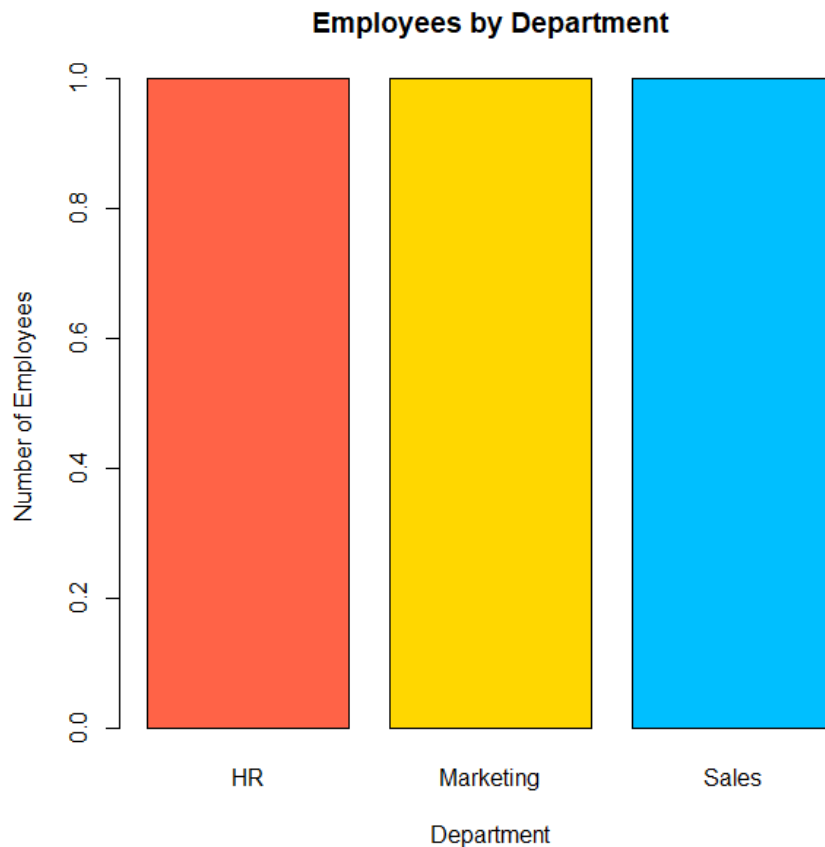
```
legend("bottomright",  
      legend="Performance",  
      col="blue",  
      pch=16,  
      lty=1)
```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
dept <- table(employee$Department)
```

```
barplot(dept,  
        col=c("tomato","gold","deepskyblue"),  
        border="black",  
        xlab="Department",  
        ylab="Number of Employees",  
        main="Employees by Department")
```



3. Build a table to display the performance data for each employee. Label the table elements.

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
```

```
employee$Grade <- ifelse(employee$PerformanceScore >= 90,"A",
  ifelse(employee$PerformanceScore >= 80,"B","C"))
```

```
print(employee)
```

```
print(employee)
EmployeeID Department YearsService PerformanceScore Grade
1 Sales 5 85 B
2 HR 3 92 A
3 Marketing 7 78 C
```

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

```
library(shiny)
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
employee$Grade <- ifelse(employee$PerformanceScore >= 90,"A",
  ifelse(employee$PerformanceScore >= 80,"B","C"))
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  sidebarLayout(
    sidebarPanel
    selectInput("view",
      "Select View",
      choices=c("Line Chart",
        "Bar Chart",
        "Employee Table"))
  ),
  mainPanel(
    plotOutput("plot"),
    tableOutput("table")
  )
)
)
)

server <- function(input,output){

  output$plot <- renderPlot({

    if(input$view=="Line Chart"){
```

```
plot(employee$EmployeeID,  
      employee$PerformanceScore,  
      type="o",  
      pch=16,  
      col="blue",  
      lwd=2,  
      xlab="Employee ID",  
      ylab="Performance Score",  
      main="Employee Performance Trend")
```

```
legend("bottomright",  
      legend="Performance",  
      col="blue",  
      pch=16,  
      lty=1)
```

```
}
```

```
else if(input$view=="Bar Chart"){
```

```
dept <- table(employee$Department)
```

```
barplot(dept,  
        col=c("tomato","gold","deepskyblue"),  
        border="black",  
        xlab="Department",  
        ylab="Number of Employees",  
        main="Employees by Department")
```

```
}
```

```
}}
```

```
output$table <- renderTable({
```

```
if(input$view=="Employee Table"){
```

```
employee
```

```
}
```

```
}}
```

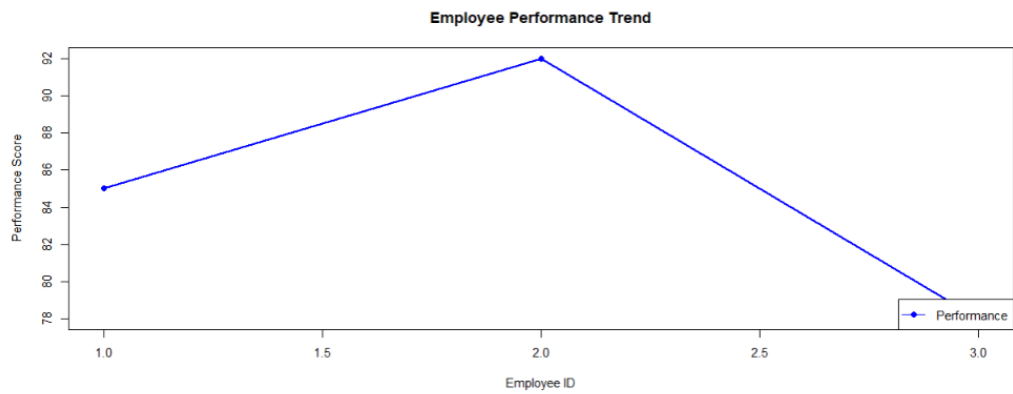
```
}
```

```
shinyApp(ui=ui,server=server)
```

Employee Performance Dashboard

Select View

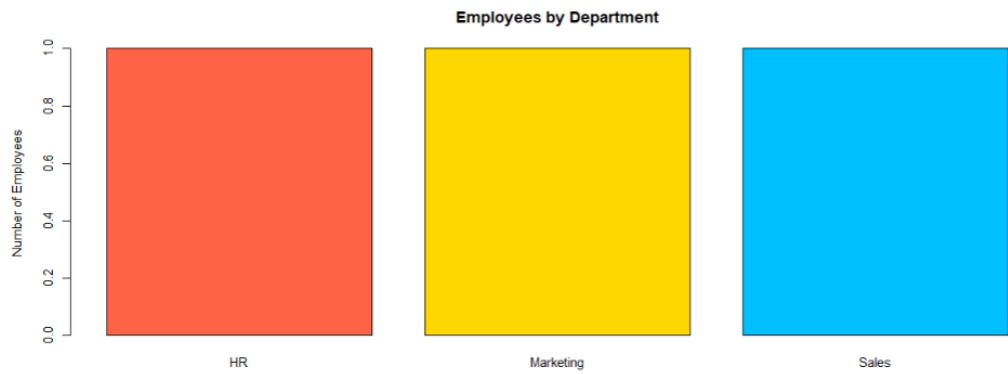
Line Chart



Employee Performance Dashboard

Select View

Bar Chart



Employee Performance Dashboard

Select View

Employee Table

EmployeeID	Department	YearsService	PerformanceScore	Grade
1.00	Sales	5.00	85.00	B
2.00	HR	3.00	92.00	A
3.00	Marketing	7.00	78.00	C

SET 9

Dataset: Online_Learning_Activity.csv

```
data <- data.frame(  
  Student_ID = c("L01","L02","L03","L04","L05","L06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Course = c("R","R","SQL","R","R","SQL"),  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched = c(12,15,8,18,9,14),  
  Quiz_Score = c(78,85,65,92,70,88),  
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",  
    "2025-02-08","2025-03-12","2025-03-12")  
)
```

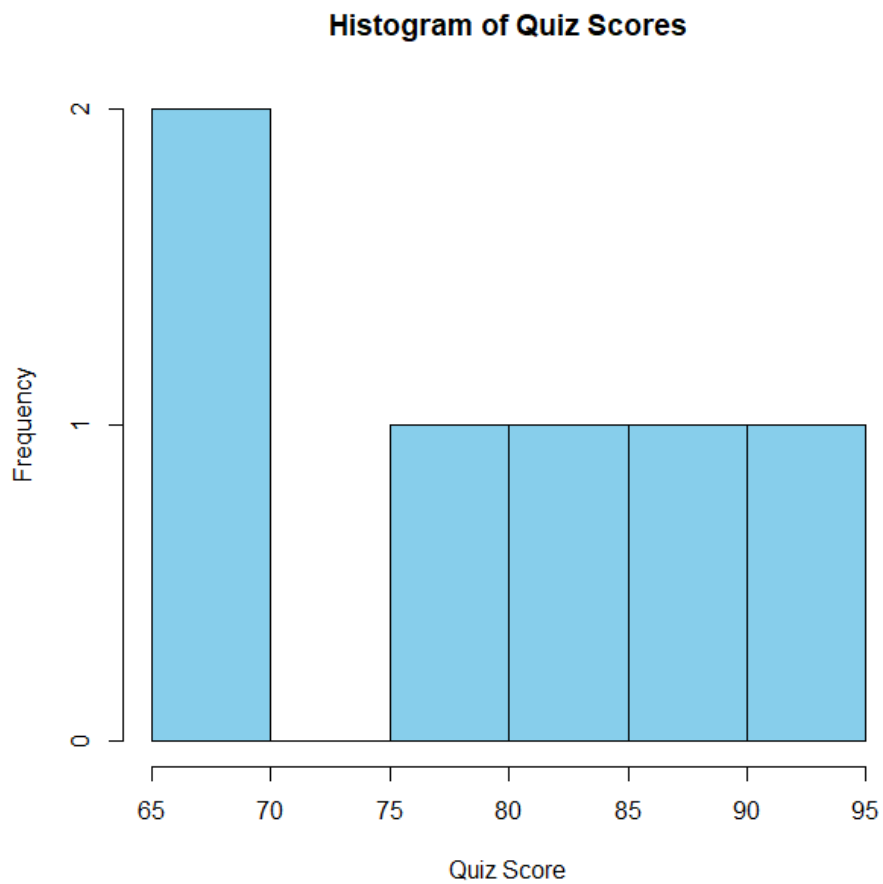
Data

```
> data  
  Student_ID Gender Age Course Study_Time Videos_Watched Quiz_Score Login_Date  
1      L01   Male  20     R        3.5           12           78 2025-01-05  
2      L02 Female  22     R        4.2           15           85 2025-01-05  
3      L03   Male  19    SQL        2.0            8           65 2025-02-08  
4      L04 Female  21     R        5.0           18           92 2025-02-08  
5      L05   Male  23     R        2.5            9           70 2025-03-12  
6      L06 Female  20    SQL        4.0           14           88 2025-03-12  
> |
```

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

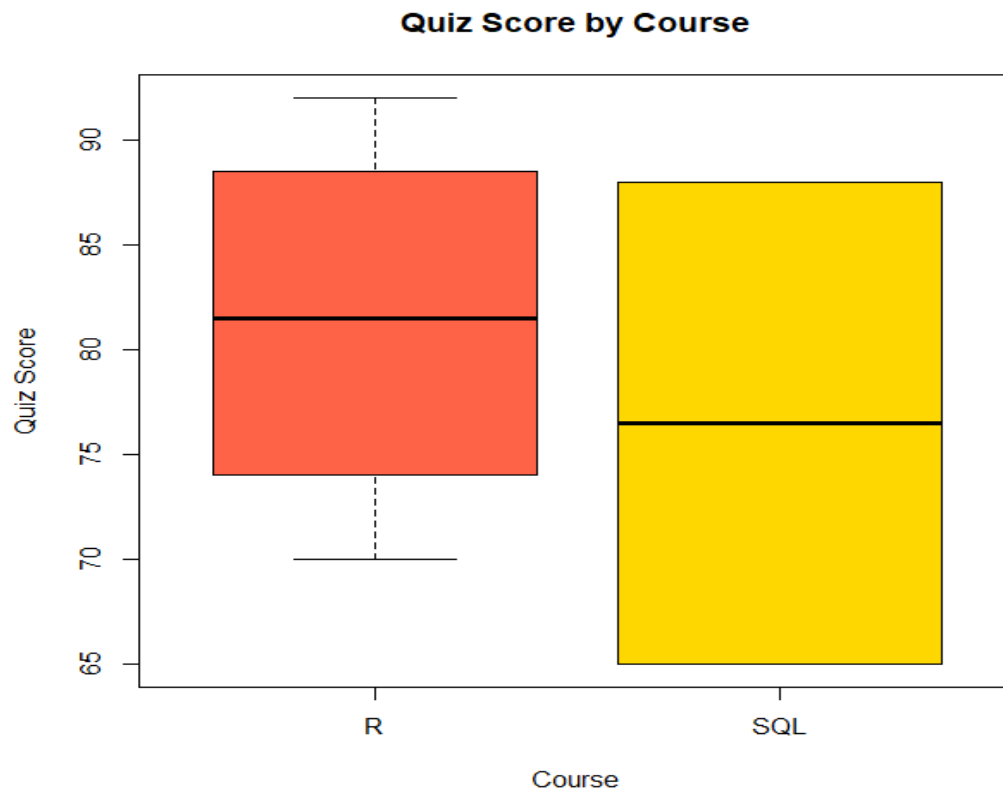
Histogram of quiz scores:

```
hist(data$Quiz_Score,  
  col="skyblue",  
  border="black",  
  main="Histogram of Quiz Scores",  
  xlab="Quiz Score",  
  ylab="Frequency")
```



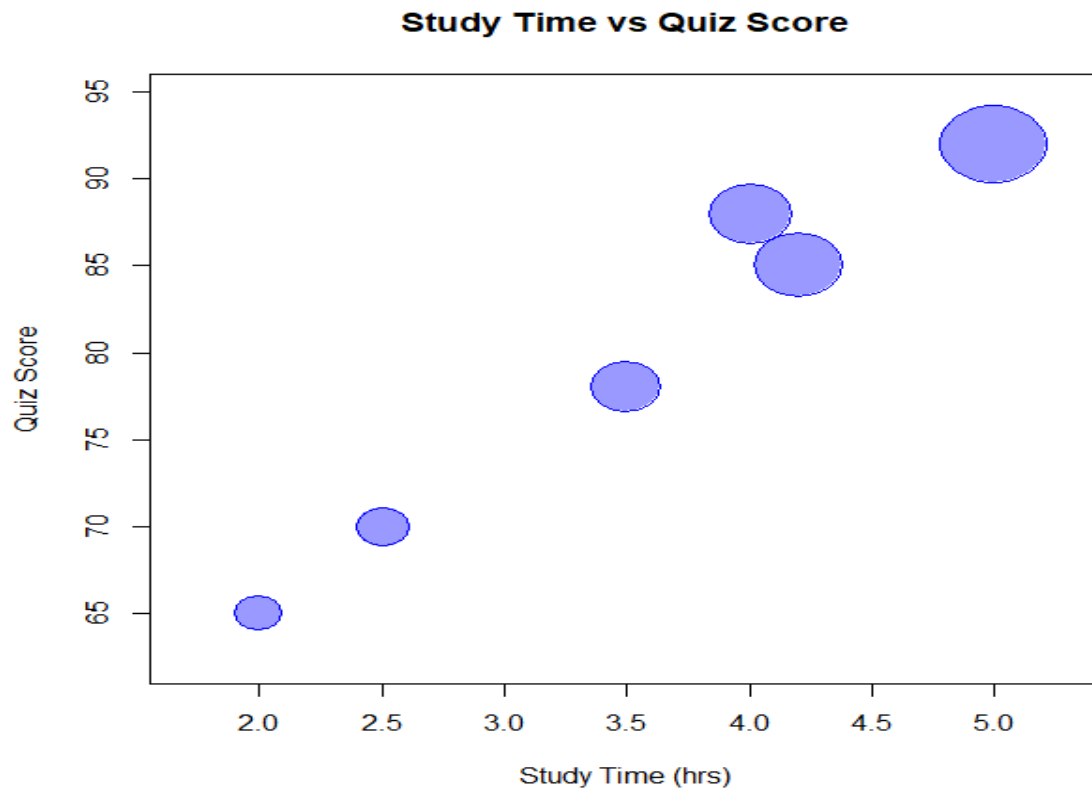
Boxplot of quiz scores:

```
boxplot(Quiz_Score~Course,  
        data=data,  
        col=c("tomato","gold"),  
        main="Quiz Score by Course",  
        xlab="Course",  
        ylab="Quiz Score")
```

2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
symbols(data$Study_Time,  
        data$Quiz_Score,  
        circles=data$Videos_Watched,  
        inches=0.3,  
        bg=rgb(0,0,1,0.4),  
        fg="blue",  
        xlab="Study Time (hrs)",  
        ylab="Quiz Score",  
        main="Study Time vs Quiz Score")
```



- Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

Convert Date and Calculate Monthly Average:

```
data$Login_Date <- as.Date(data$Login_Date)
data$Month <- format(data$Login_Date,"%Y-%m")
avg <- aggregate(Quiz_Score~Month,
                 data=data,
                 mean)
print(avg)
```

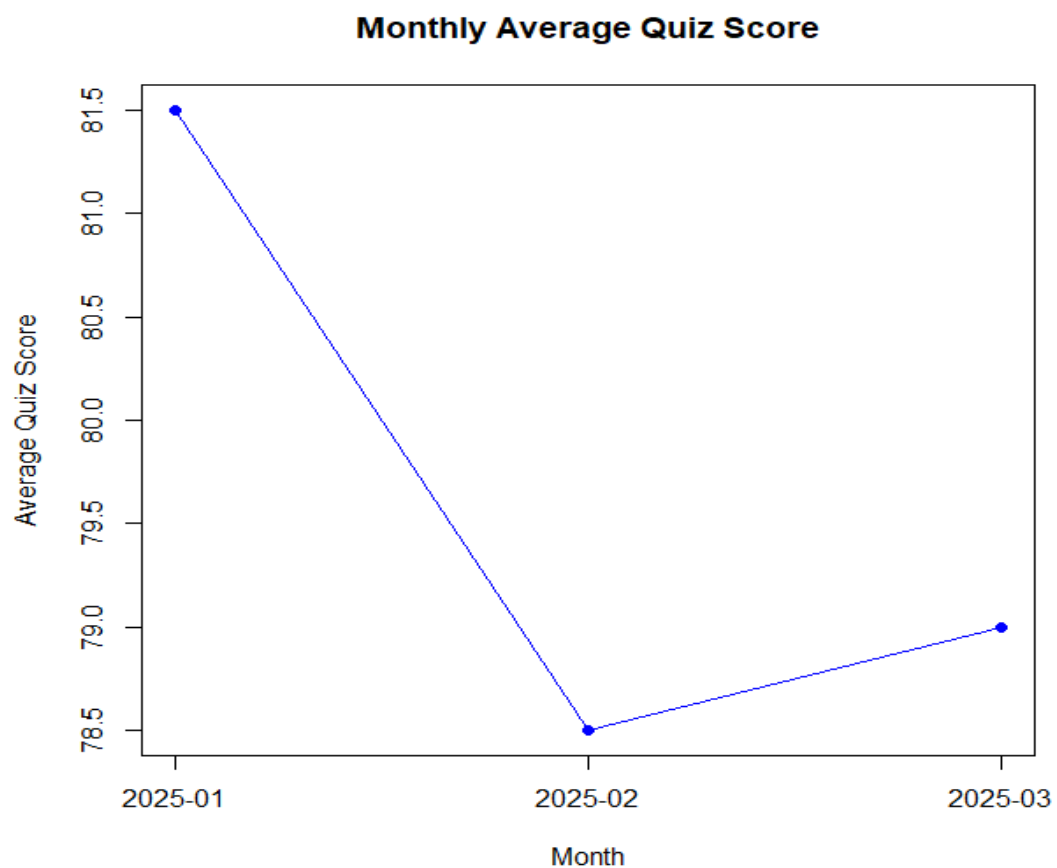
```
> print(avg)
      Month Quiz_Score
1 2025-01         81.5
2 2025-02         78.5
3 2025-03         79.0
> |
```

Line Chart

```
plot(avg$Quiz_Score,
     type="o",
```

```
pch=16,  
col="blue",  
xaxt="n",  
xlab="Month",  
ylab="Average Quiz Score",  
main="Monthly Average Quiz Score")
```

```
axis(1,  
  at=1:nrow(avg),  
  labels=avg$Month)
```



Moving Average:

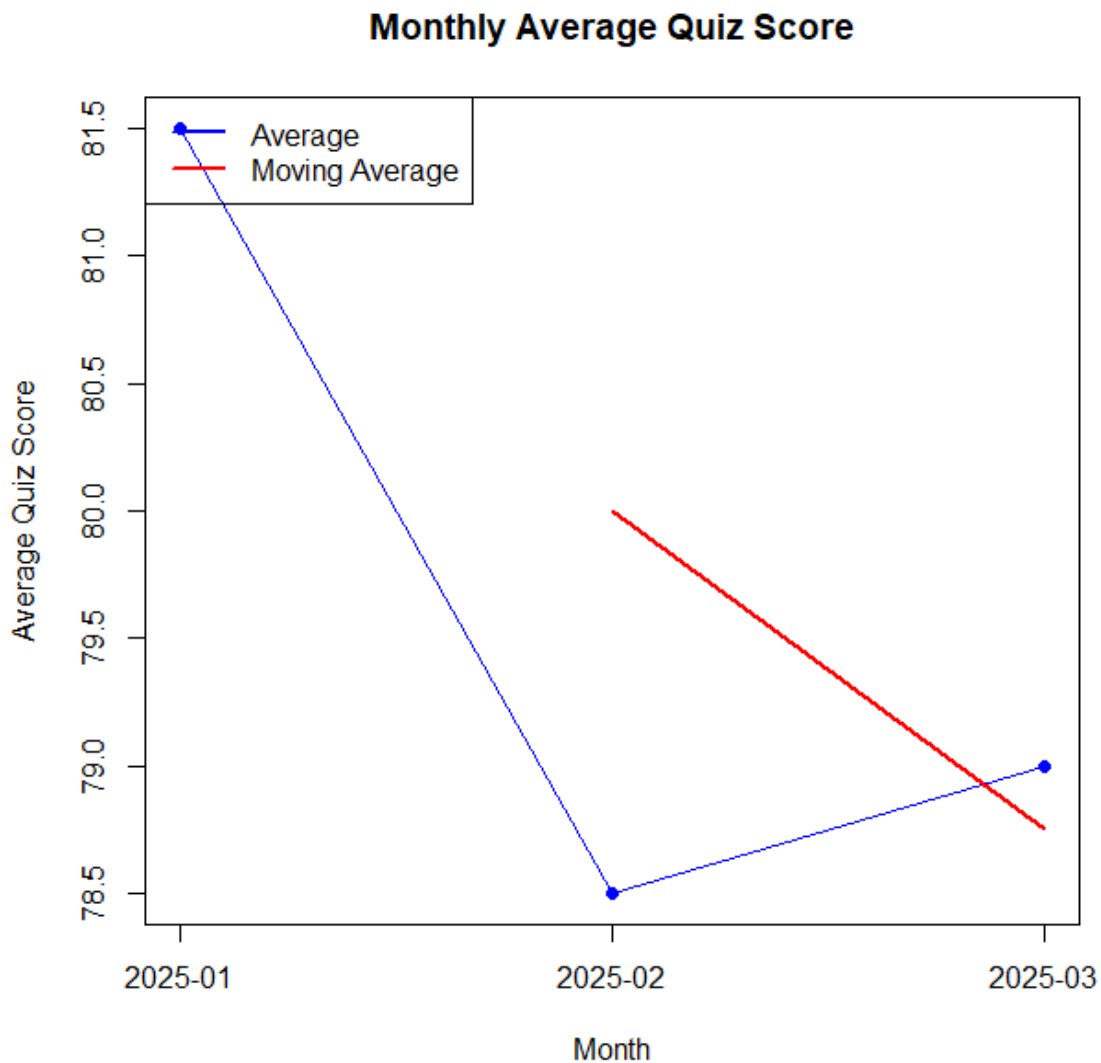
```
ma <- stats::filter(avg$Quiz_Score,  
  rep(1/2,2),  
  sides=1)
```

```
lines(ma,  
  col="red",  
  lwd=2)
```

```

legend("topleft",
      legend=c("Average","Moving Average"),
      col=c("blue","red"),
      lty=1,
      lwd=2)

```



4. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

```
library(shiny)
```

```

data <- data.frame(
  Student_ID=c("L01","L02","L03","L04","L05","L06"),
  Gender=c("Male","Female","Male","Female","Male","Female"),
  Age=c(20,22,19,21,23,20),
  Course=c("R","R","SQL","R","R","SQL"),

```

```

Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),
Videos_Watched=c(12,15,8,18,9,14),
Quiz_Score=c(78,85,65,92,70,88)
)

ui <- fluidPage(

titlePanel("Online Learning Activity Dashboard"),

sidebarLayout(

sidebarPanel(

selectInput("view",
            "Select View",
            choices=c("Bar Chart",
                      "Scatter Plot",
                      "Student Table"))

),

mainPanel(

plotOutput("plot"),
tableOutput("table")

)

)

)

server <- function(input,output){

output$plot <- renderPlot({

if(input$view=="Bar Chart"){

avg <- aggregate(Quiz_Score~Course,
                  data=data,
                  mean)

```

```
barplot(avg$Quiz_Score,
        names.arg=avg$Course,
        col=c("tomato","gold"),
        border="black",
        xlab="Course",
        ylab="Average Quiz Score",
        main="Average Quiz Score by Course")
```

```
}
```

```
else if(input$view=="Scatter Plot"){
```

```
symbols(data$Study_Time,
        data$Quiz_Score,
        circles=data$Videos_Watched,
        inches=0.3,
        bg=rgb(0,0,1,0.4),
        fg="blue",
        xlab="Study Time (hrs)",
        ylab="Quiz Score",
        main="Study Time vs Quiz Score")
```

```
}
```

```
})
```

```
output$table <- renderTable({
```

```
if(input$view=="Student Table"){
```

```
data
```

```
}
```

```
})
```

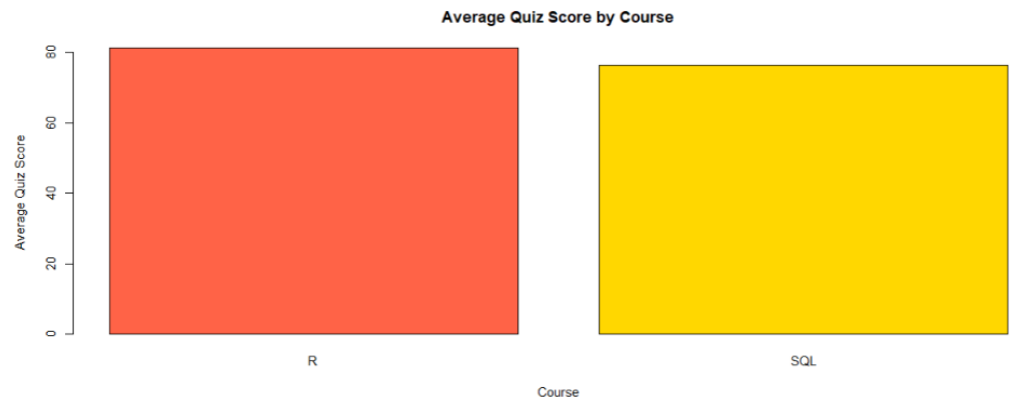
```
}
```

```
shinyApp(ui=ui,server=server)
```

Online Learning Activity Dashboard

Select View

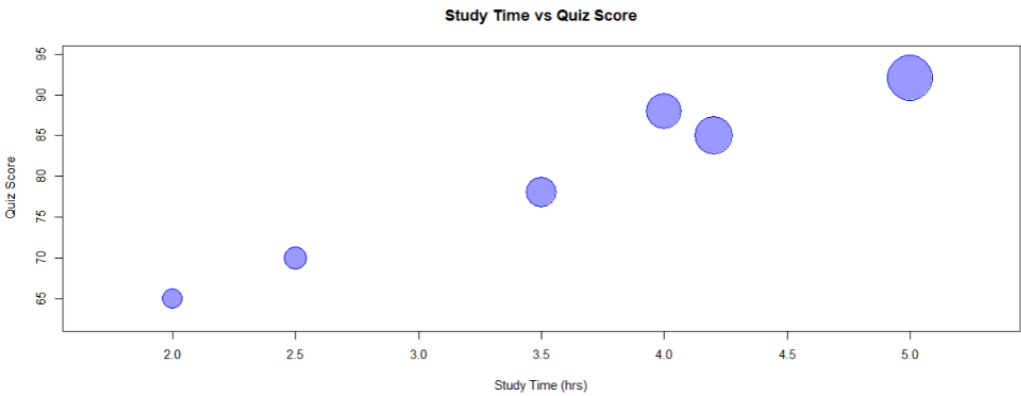
Bar Chart



Online Learning Activity Dashboard

Select View

Scatter Plot



Online Learning Activity Dashboard

Select View

Student Table

Student_ID	Gender	Age	Course	Study_Time	Videos_Watched	Quiz_Score
L01	Male	20.00	R	3.50	12.00	78.00
L02	Female	22.00	R	4.20	15.00	85.00
L03	Male	19.00	SQL	2.00	8.00	65.00
L04	Female	21.00	R	5.00	18.00	92.00
L05	Male	23.00	R	2.50	9.00	70.00
L06	Female	20.00	SQL	4.00	14.00	88.00

SET 10 SURVEY RESPONSES ANALYSIS

Dataset: Survey Responses

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

Survey

```
> survey  
  SurveyID Question1 Question2 Question3  
1         1         A         B         C  
2         2         B         A         D  
3         3         C         A         B  
> |
```

1. Create a grouped bar chart to visualize the distribution of answers for Question 1.
Label the chart elements.

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
q1 <- table(survey$Question1)
```

```
barplot(q1,  
  beside=TRUE,  
  col=c("tomato","gold","deepskyblue"),  
  border="black",  
  main="Distribution of Question 1 Responses",  
  xlab="Responses",  
  ylab="Frequency")
```

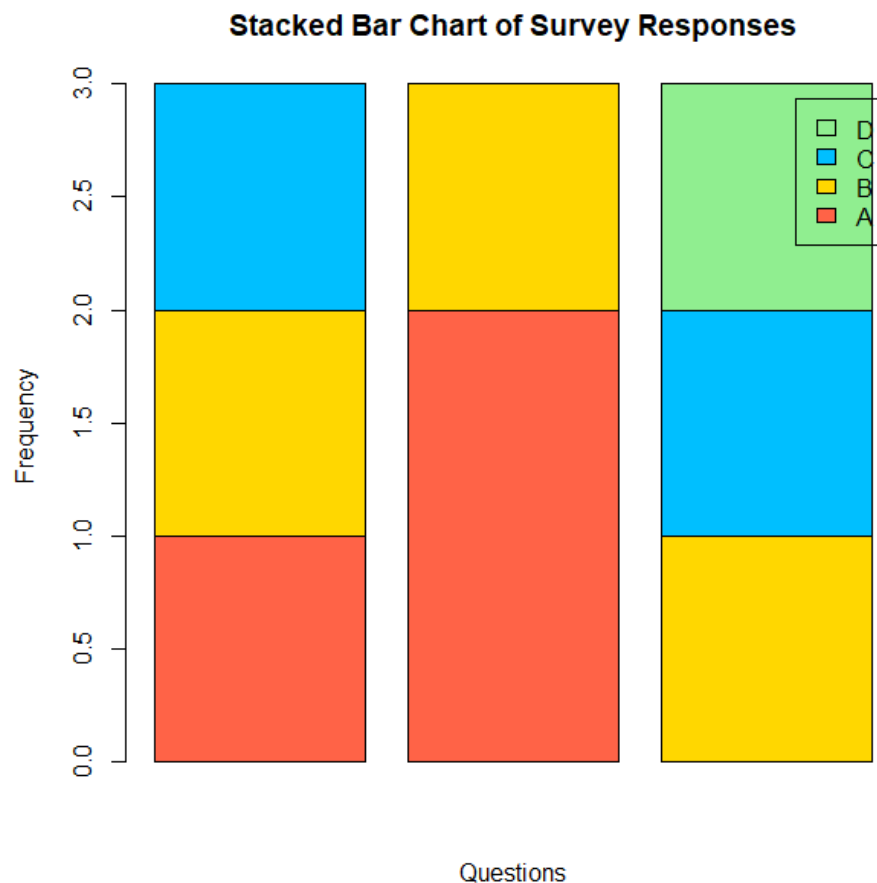



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

response_table <- rbind(
  table(factor(survey$Question1, levels=c("A","B","C","D"))),
  table(factor(survey$Question2, levels=c("A","B","C","D"))),
  table(factor(survey$Question3, levels=c("A","B","C","D")))
)

barplot(t(response_table),
  beside=FALSE,
  col=c("tomato","gold","deepskyblue","lightgreen"),
  legend.text=c("A","B","C","D"),
  xlab="Questions",
  ylab="Frequency",
  main="Stacked Bar Chart of Survey Responses")
```



3. Build a table to show the survey response data for each survey. Label the table elements.

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
```

```
survey$TotalResponses <- 3
```

```
print(survey)
```

```
print(survey)
  SurveyID Question1 Question2 Question3 TotalResponses
1        1         A         B         C              3
2        2         B         A         D              3
3        3         C         A         B              3
```

4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

```
library(shiny)

survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

ui <- fluidPage(

  titlePanel("Survey Responses Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("view",
        "Select View",
        choices=c("Grouped Bar Chart",
          "Stacked Bar Chart",
          "Survey Table"))

    ),

    mainPanel(

      plotOutput("plot"),
      tableOutput("table")

    )

  )

)

server <- function(input,output){
```

```

output$plot <- renderPlot({

if(input$view=="Grouped Bar Chart"){

q1 <- table(survey$Question1)

barplot(q1,
        col=c("tomato","gold","deepskyblue"),
        border="black",
        main="Question 1 Responses",
        xlab="Responses",
        ylab="Frequency")

}

else if(input$view=="Stacked Bar Chart"){

response_table <- rbind(
table(factor(survey$Question1,levels=c("A","B","C","D"))),
table(factor(survey$Question2,levels=c("A","B","C","D"))),
table(factor(survey$Question3,levels=c("A","B","C","D")))
)

barplot(t(response_table),
        beside=FALSE,
        col=c("tomato","gold","deepskyblue","lightgreen"),
        legend.text=c("A","B","C","D"),
        xlab="Questions",
        ylab="Frequency",
        main="Survey Responses")

}

})

output$table <- renderTable({

if(input$view=="Survey Table"){

survey

```

}

})

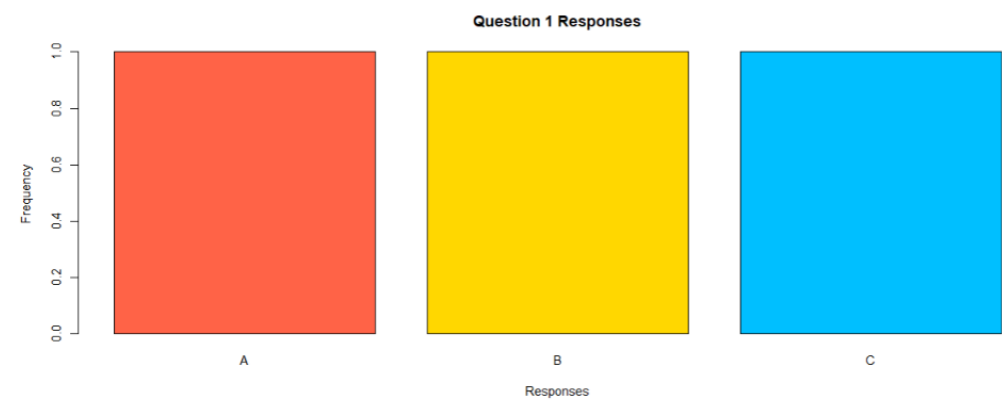
}

shinyApp(ui=ui,server=server)

Survey Responses Dashboard

Select View

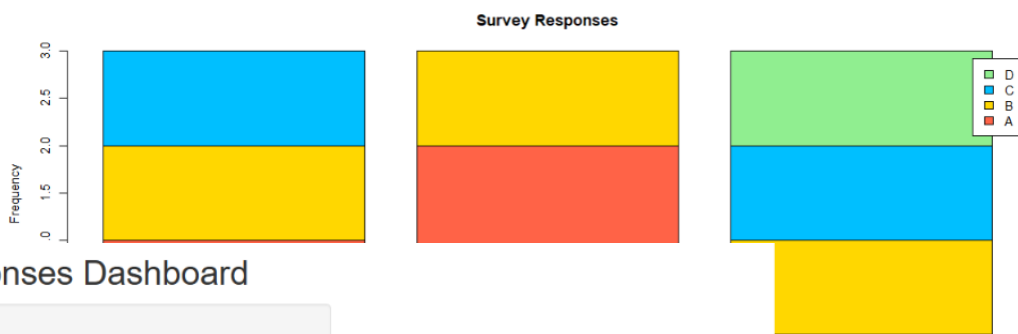
Grouped Bar Chart



Survey Responses Dashboard

Select View

Stacked Bar Chart



Survey Responses Dashboard

Select View

Survey Table

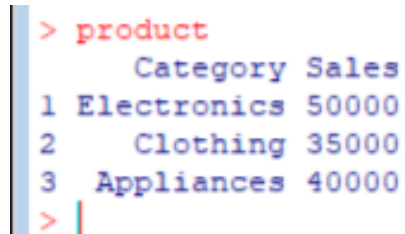
SurveyID	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

SET 11: PRODUCT CATEGORY ANALYSIS

Dataset: Product Categories

```
product <- data.frame(  
  Category = c("Electronics","Clothing","Appliances"),  
  Sales = c(50000,35000,40000)  
)
```

Product



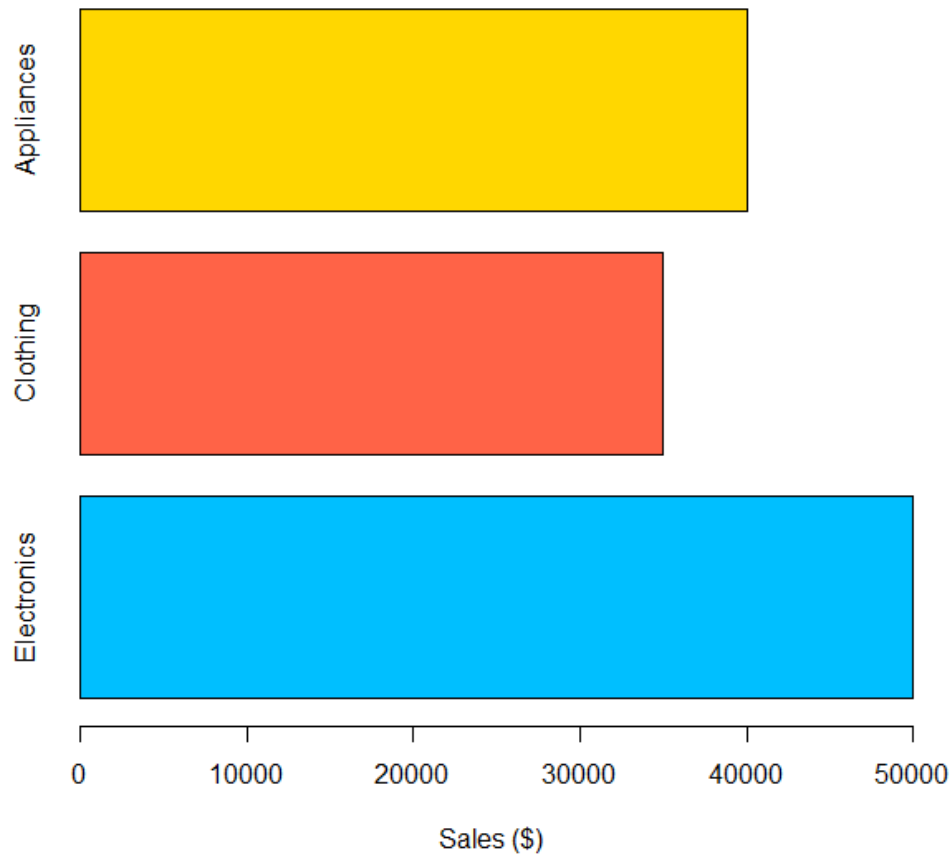
```
> product  
      Category Sales  
1 Electronics 50000  
2   Clothing 35000  
3 Appliances 40000  
> |
```

1.Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
product <- data.frame(  
  Category = c("Electronics","Clothing","Appliances"),  
  Sales = c(50000,35000,40000)  
)
```

```
barplot(product$Sales,  
  horiz=TRUE,  
  names.arg=product$Category,  
  col=c("deepskyblue","tomato","gold"),  
  border="black",  
  xlab="Sales ($)",  
  main="Funnel Chart for Product Category Sales")
```

Funnel Chart for Product Category Sales



2. Build a table to display the sales data for each product category. Label the table elements.

```
product <- data.frame(  
  Category=c("Electronics","Clothing","Appliances"),  
  Sales=c(50000,35000,40000)  
)
```

```
product$Sales_in_Lakhs <- product$Sales/100000
```

```
print(product)
```

```
> print(product)
      Category Sales Sales_in_Lakhs
1 Electronics 50000          0.50
2   Clothing 35000          0.35
3  Appliances 40000          0.40
> |
```

3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

```
library(shiny)
```

```
product <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Product Category Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput("view",
        "Select View",
        choices=c("Pie Chart",
          "Funnel Chart",
          "Sales Table"))
    ),
```

```
    mainPanel(
```

```
      plotOutput("plot"),
```

```
      tableOutput("table")
    )
  )
)
```



```

)

server <- function(input,output){

output$plot <- renderPlot({

if(input$view=="Pie Chart"){

pie(product$Sales,
  labels=product$Category,
  col=c("deepskyblue","tomato","gold"),
  main="Sales Distribution")

}

else if(input$view=="Funnel Chart"){

barplot(product$Sales,
  horiz=TRUE,
  names.arg=product$Category,
  col=c("deepskyblue","tomato","gold"),
  border="black",
  xlab="Sales",
  main="Funnel Chart")

}

})

output$table <- renderTable({

if(input$view=="Sales Table"){

product

}

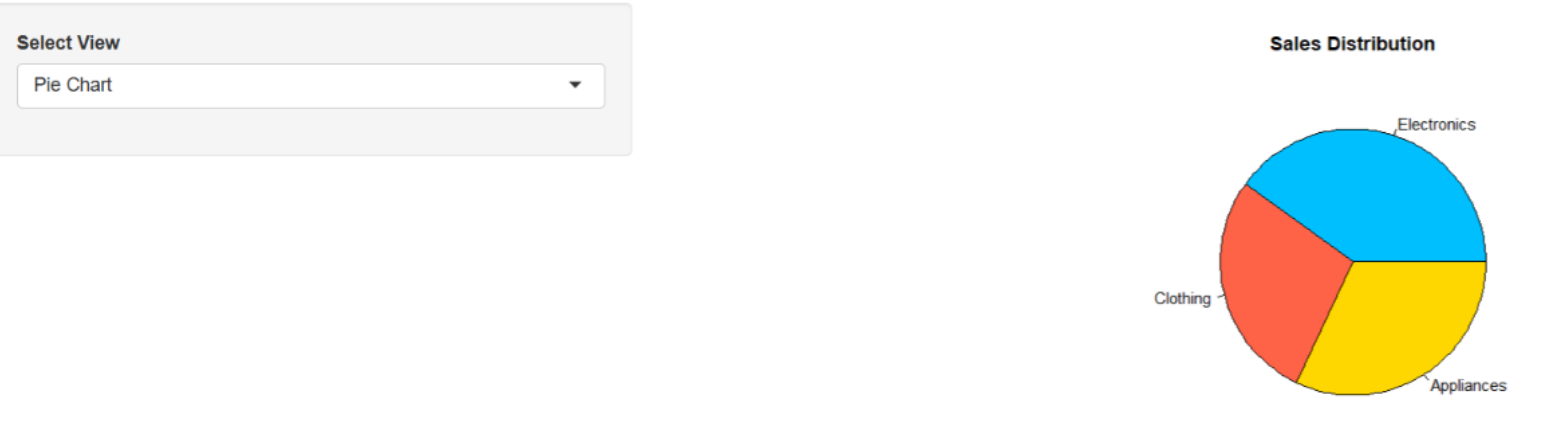
})

}

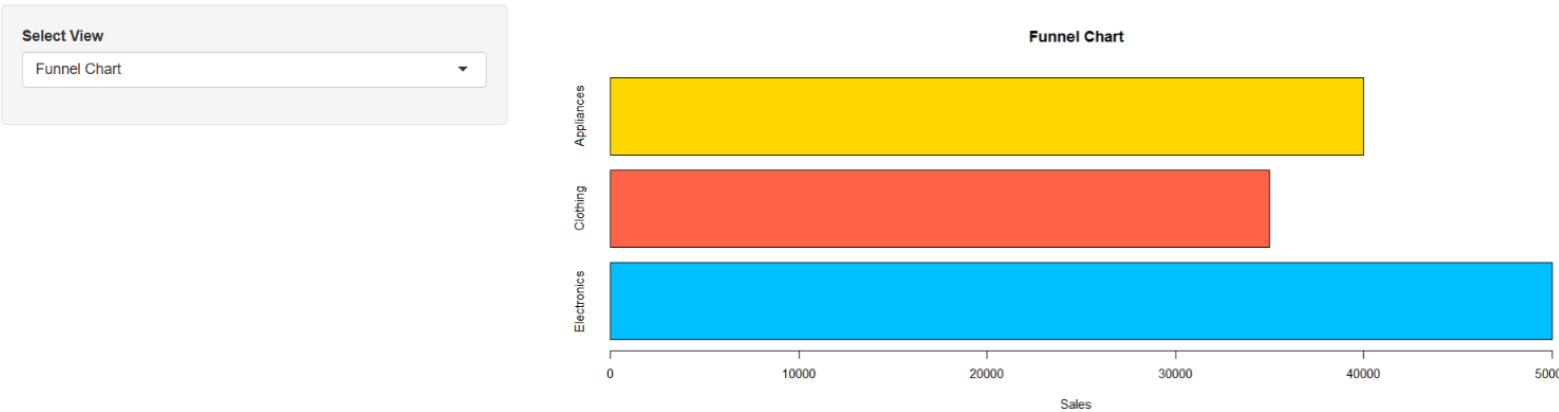
```

```
shinyApp(ui,server)
```

Product Category Dashboard



Product Category Dashboard



Product Category Dashboard

Select View

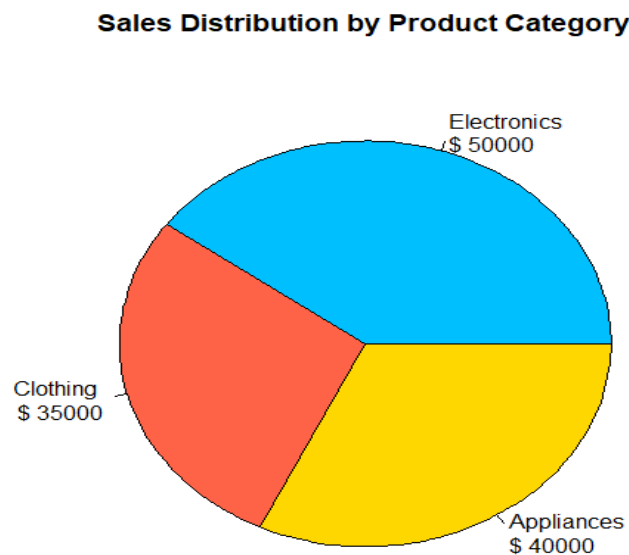
Sales Table

Category	Sales
Electronics	50000.00
Clothing	35000.00
Appliances	40000.00

```
4.Create a pie chart to represent the distribution of sales across product categories.
Include labels.
product <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
```

)

```
pie(product$Sales,  
     labels=paste(product$Category,"\\n$",product$Sales),  
     col=c("deepskyblue","tomato","gold"),  
  
     main="Sales Distribution by Product Category")
```



SET 12

Dataset: Website Traffic

```
traffic <- data.frame(  
  Date=c("2023-01-01","2023-01-02","2023-01-03"),  
  PageViews=c(1500,1600,1400),  
  CTR=c(2.3,2.7,2.0)  
)
```

Traffic

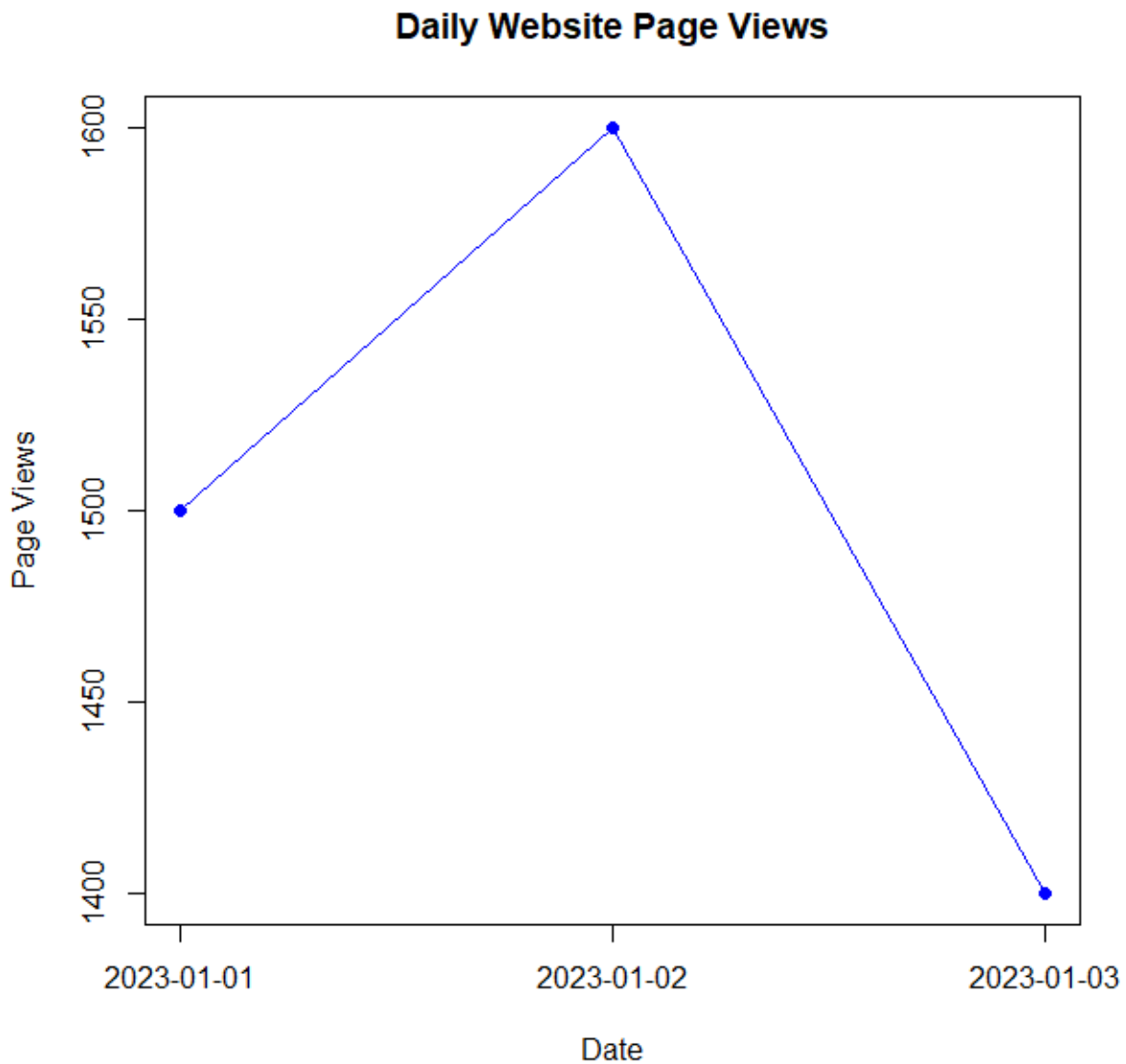
```
> traffic
      Date PageViews CTR
1 2023-01-01      1500 2.3
2 2023-01-02      1600 2.7
3 2023-01-03      1400 2.0
> |
```

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

```
plot(traffic$PageViews,
     type="o",
     pch=16,
     col="blue",
     xaxt="n",
     xlab="Date",
     ylab="Page Views",
     main="Daily Website Page Views")
```

```
axis(1,
     at=1:3,
```

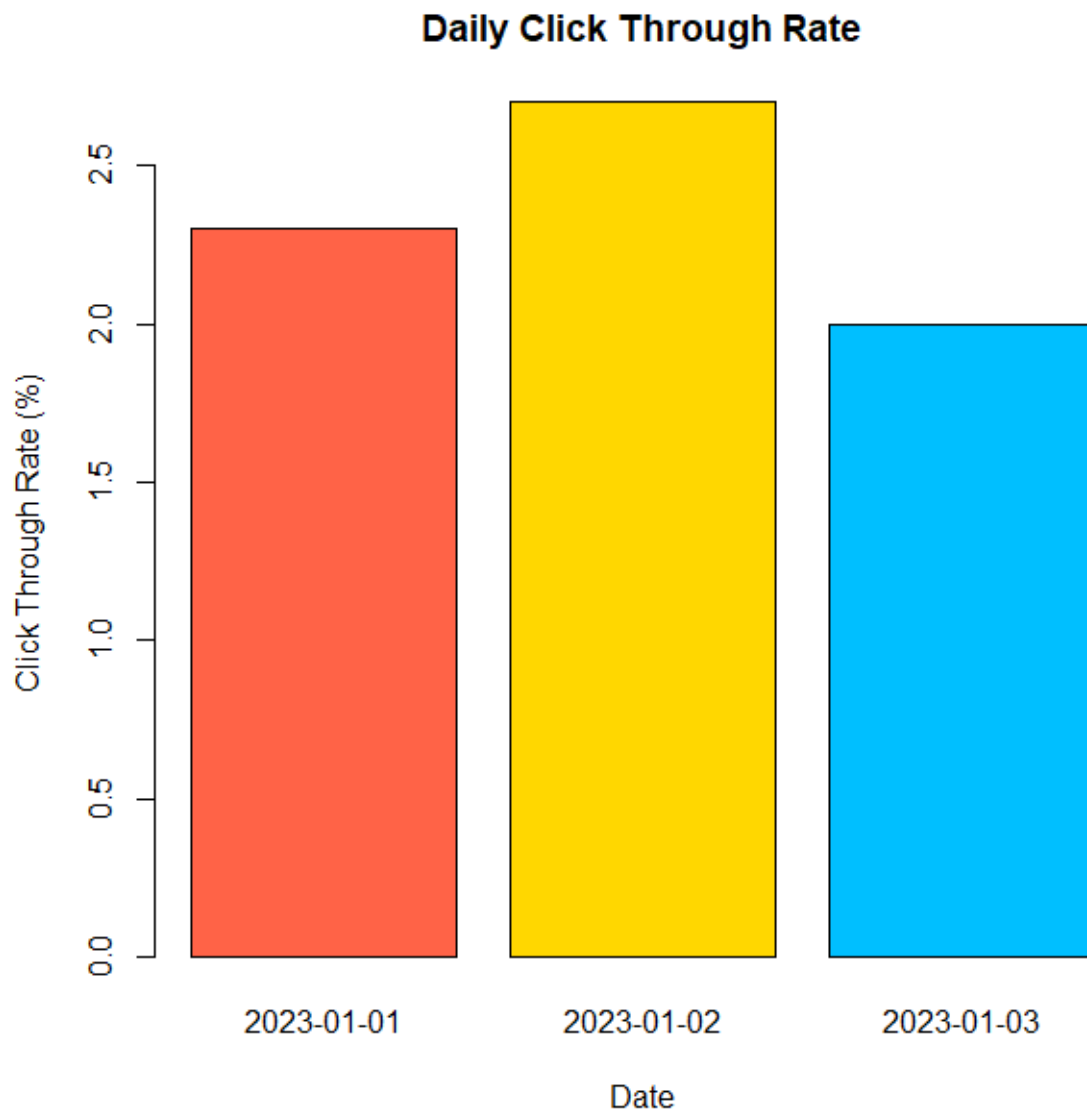
```
labels=traffic$Date)
```



2. **Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.**

```
barplot(traffic$CTR,  
        names.arg=traffic$Date,  
        col=c("tomato","gold","deepskyblue"),  
        border="black",  
        xlab="Date",  
        ylab="Click Through Rate (%)",
```

```
main="Daily Click Through Rate")
```



3. Build a table to display the geographic data for each city. Label the table elements.

```
traffic <- data.frame(  
  Date=c("2023-01-01","2023-01-02","2023-01-03"),  
  PageViews=c(1500,1600,1400),  
  CTR=c(2.3,2.7,2.0)  
)  
traffic$EstimatedClicks <- round(traffic$PageViews * traffic$CTR / 100)  
print(traffic)
```

```
> print(traffic)
      Date PageViews CTR EstimatedClicks
1 2023-01-01      1500 2.3             34
2 2023-01-02      1600 2.7             43
3 2023-01-03      1400 2.0             28
> |
```

4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

```
library(shiny)
```

```
traffic <- data.frame(
  Date=c("2023-01-01","2023-01-02","2023-01-03"),
  PageViews=c(1500,1600,1400),
  CTR=c(2.3,2.7,2.0)
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Website Traffic Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput("view",
        "Select View",
        choices=c("Line Chart",
          "Bar Chart",
          "Traffic Table"))
```

```
    ),
```

```
mainPanel(  
  
  plotOutput("plot"),  
  
  tableOutput("table")  
  
)  
  
)  
  
)  
  
server <- function(input,output){  
  
  output$plot <- renderPlot({  
  
    if(input$view=="Line Chart"){  
  
      plot(traffic$PageViews,  
        type="o",  
        pch=16,  
        col="blue",  
        xaxt="n",  
        xlab="Date",  
        ylab="Page Views",  
        main="Daily Website Page Views")  
    }  
  })  
}
```



```

axis(1,
     at=1:3,
     labels=traffic$Date)

}

else if(input$view=="Bar Chart"){

barplot(traffic$CTR,
        names.arg=traffic$Date,
        col=c("tomato","gold","deepskyblue"),
        border="black",
        xlab="Date",
        ylab="CTR (%)",
        main="Daily Click Through Rate")

}

})

output$table <- renderTable({

if(input$view=="Traffic Table"){

traffic

}

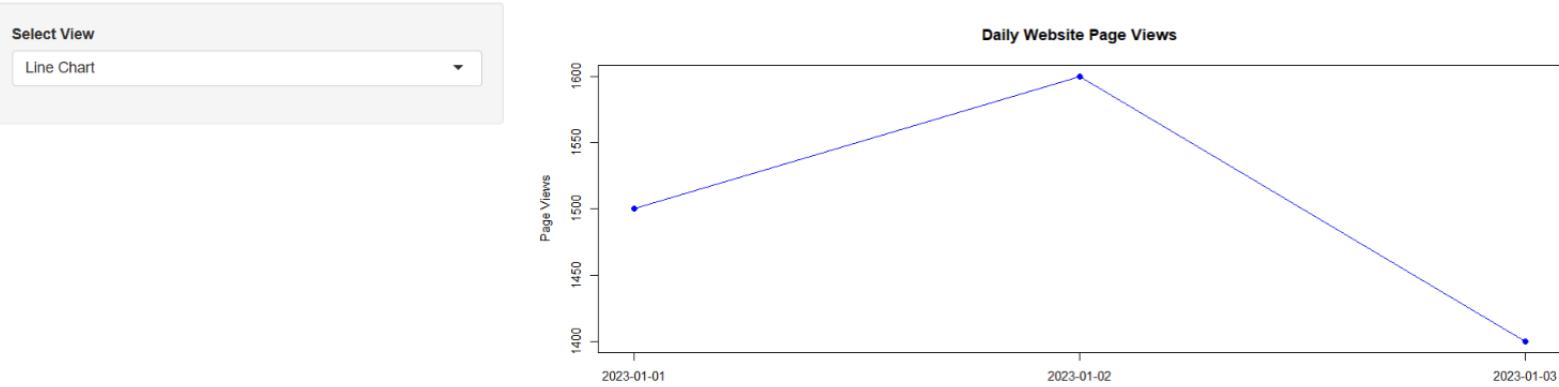
})

}

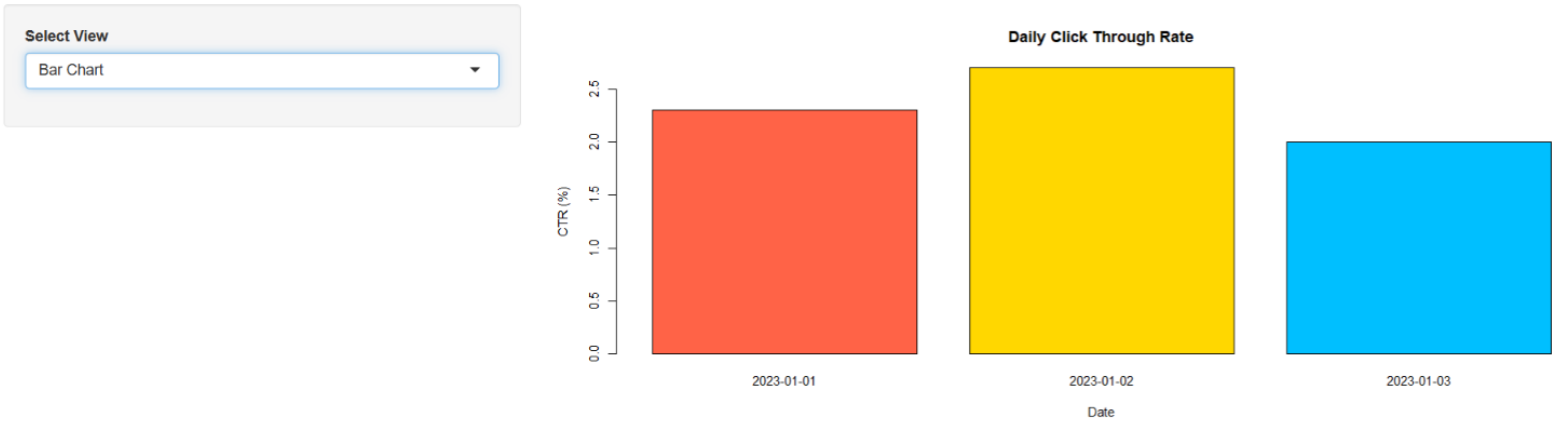
```

shinyApp(ui,server)

Website Traffic Dashboard



Website Traffic Dashboard



Website Traffic Dashboard

Select View

Traffic Table

Date	PageViews	CTR
2023-01-01	1500.00	2.30
2023-01-02	1600.00	2.70
2023-01-03	1400.00	2.00